50. 5th Comp

POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2025
Programme: BE | Full Marks : 100
Course: Image Processing and Pattern Recognition (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define digital image processing with its representation. Explain its application in real life. 8

   b) Perform the histogram Stretching on the image shown below. 7

Gary level	0	1	2	3	4	5	6	7
Frequency	0	50	60	80	40	80	10	0

2. a) Explain the step of filtering in frequency domain. Describe the Mean and median filters for image enhancement. 8

   b) What do you mean by histogram modelling explain with its basic properties. Equalize a histogram of a 3-bit 5*5 image as given below. Output the histogram equalized image. 7

4	3	3	4	4
3	4	5	4	5
3	5	5	5	3
3	4	5	4	3
4	4	5	4	4

OR

Given the following gray level histogram of an image. Compute the gray levels after histogram equalization.

M	0	1	2	3	4	5	6	7
Gray Level	100	300	400	500	600	20	180	400

3. a) Define DFT Transform with its expressions? List out its properties and uses. 8

   b) Why do we need compression? Compress the string "ABCDABCEABCDBCD" using LZW coding. 7

01230 1 240 123123

4. a) Explain Image Restoration and Degradation model with necessary diagrams. 8

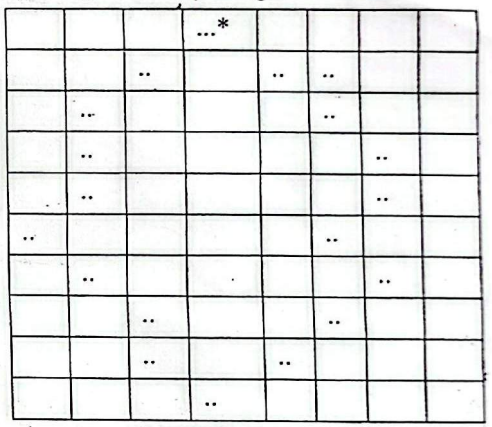
   b) Define Image Compression. Explain Lossless and Lossy Image Compression Technique. 7

5. a) Why image segmentation is necessary in image processing? Explain segmentation by region splitting and merging algorithm. 8

OR

List different image segmentation methods. Describe the region growing algorithm with steps and suitable example.

   b) Define chain code and shape number. Also find the Shape Number for the following Boundary Image. 7



6. a) Describe the steps of pattern recognition with block diagram. 7

   b) Define PCA. Explain how features of an image are extracted using PCA? 8

7. Write short notes on: **(Any two)** 2×5

   a) Elements of information theory

   b) Fourier descriptor

   c) Hopefield Network

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POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2025
Programme: BE | Full Marks : 100
Course: Image Processing and Pattern Recognition (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
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Attempt all the questions.

1. a) Define sampling and quantization in image acquisition. How adjacency of the pixels affects connectivity and path between the pixels? Explain with suitable examples. 8

   b) List the basic gray level transformation technique. Perform the contrast stretching on the image given below. 7

Gray level	0	1	2	3	4	5	6	7
Frequency	0	40	60	80	50	90	10	0

2. a) What is histograms equalization? Perform the histogram equalization on the following 8-bit gray level image. 8

3	2	4	5
1	2	2	2
7	3	1	2
7	6	4	5

OR

A grayscale image has poor contrast. Describe how histogram equalization improves the image quality. Illustrate with a sample histogram transformation.

   b) Explain different noise models in an image with the graphs and their PDF. 7

3. a) Define DFT Transform with its expressions? List out its properties and uses. 8

OR

Derive the Kernel for Haar transform of N=2 order.

5	0	1	1
0	4	2	4
5	5	7	4
3	1	0	1

   b) Define thresholding. How can we select the efficient threshold value for image segmentation? Mention limitations of single valued thresholding for global thresholding? How adaptive thresholding solve this issue? Explain. 7

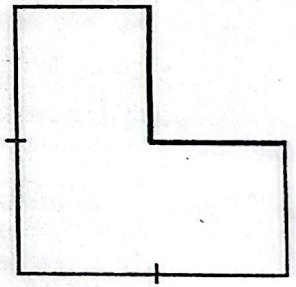
4. a) Define Image Compression. Explain Lossless predictive coding with necessary diagram. 8

   b) What is edge detection? Explain the edge detection operators with mask. 7

5. a) Explain Huffman coding. Determine Huffman code for given image data and calculate the efficiency. 8

Gray Levels	0	1	2	3	4	5	6	7
Frequency	7	13	23	44	58	28	16	11

   b) Define shape number. How does the order of a descriptor determine the efficiency of shape representation? Demonstrate that shape numbers are origin-invariant with the help of given shape. 7



6. a) Define pattern and pattern recognition system. explain the steps use in pattern recognition. 8

   b) How SIFT keypoints can be used for feature extraction and classification? Explain. 7

7. Write short notes on: **(Any two)** 2×5
   a) Discrete cosine transform
   b) Image degradation/restoration model
   c) Hopfield Network

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POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2025
Programme: BE | Full Marks : 100
Course: Machine Learning (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Machine Learning. Describe the typical Machine Learning workflow. 7
   b) How does regularization affect the bias-variance trade-off? Explain the Ridge and Lasso regression with its mathematical explanation. 8
2. a) What is SVM? How does the SVM classifier work? Also explain the importance of kernel functions in SVM. 7
   b) An analyst wants to build a multiple linear regression model to predict a car's price (in Rs. 100000) based on two factors: engine size (in liters) and car age (in years) 8

Car	Engine Size (X_1)	Age (X_2)	Price
1	2	5	20
2	2.2	4	22
3	1.8	6	18
4	2.5	3	24

   Fit a multiple linear regression model using matrix method.
3. a) What is market basket analysis? Using Apriori algorithm, find the frequent item sets and association rules (given minimum support count = 2, minimum confidence = 0.7) from below dataset: 7

TID	Items
100	1, 2
101	1, 3,4
102	1, 3, 4, 5
103	1, 2, 3,4

   b) Use K(=2)-Means algorithm over the data (2, 3), (3, 3), (3, 4), (5, 4), (6, 4), (6, 5), (8, 5), (9, 6) up to two iterations and show the clusters. Choose Centroid 1: (3, 3) and Centroid 2: (8, 5) as initial centroids. 8

OR

For the given data points, calculate the distance matrix and create a dendrogram using single linkage technique.

Points	X	Y
P1	1	4
P2	1	2
P3	0	4
P4	6	1
P5	7	0

4. a) What is dimensionality reduction? Explain what is PCA and the steps involved in PCA. 7
   b) Explain the Loss function in Artificial Neural Network with its mathematical equations and terminology used. 8
5. a) How multilayer perceptron uses the backpropagation and gradient descent in neural network training? Explain with an example. 7
   b) Describe the components of a Convolutional Neural Network (CNN) and explain their functions. 8

OR

Explain the architecture and functionality of Long Short-Term Memory (LSTM) networks. How do they overcome the vanishing gradient problem?

6. a) What are model validation techniques? Compare hold-out and k-fold cross-validation. 7
   b) Pokhara University recently developed a machine learning model to predict PASSED and FAILED students. While evaluating the model with a dataset of 1000 records, the system correctly predicted 480 students as PASSED and 380 students as FAILED students. However, 120 students were misclassified as FAILED students. 8
      i. Create a confusion matrix.
      ii. Compute accuracy, precision, recall, F1 score, sensitivity and specificity.
7. Write short notes on: **(Any two)** 2x5
   a) Challenges of Machine Learning
   b) Overfitting and Underfitting
   c) Hyperparameter Tuning

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POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Compiler Design

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

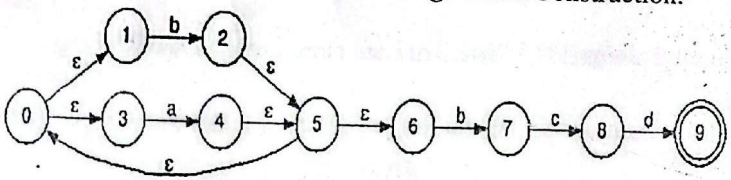
Attempt all the questions.

1. a) Define compiler. Explain the phases of a compiler with a detailed block diagram. 7

   b) Write code generation algorithm. Consider a statement x = (a-b)+(e+(c-d)) and perform simple code generation. 8

2. a) What are the functions of a lexical analyzer? Explain with suitable examples. How lexical analyzer use symbol table? 7

   b) Convert following NFA to DFA using Subset Construction. 8



3. a) Construct an **LR(1) parsing table** for the following grammar: 8

   $S' \rightarrow S$

   $S \rightarrow C\,C$

   $C \rightarrow c\,C$

   $C \rightarrow d$

OR

Consider the following grammar for arithmetic expressions:

$E \rightarrow T\,E'$

$E' \rightarrow +\,T\,E' \mid \varepsilon$

$T \rightarrow F\,T'$

$T' \rightarrow *\,F\,T' \mid \varepsilon$

$F \rightarrow (\,E\,) \mid id$

Construct the LL(1) parsing table.

   b) How does Recursive Descent Parsing works? Consider the following grammar 7

   $S \rightarrow cAd$

   $A \rightarrow ab \mid a$

   Parse the string ***cad*** using recursive descent parser.

4. a) What is the significance of symbol tables in compilers? Explain the role of hash tables in building a symbol table. 7

   b) What is the need of intermediate code optimization? Discuss the different types of Graphical IRs used in compilers. 8

5. a) Discuss the concept of Register Allocation and Assignment in code generation. 8

OR

Explain the major issues in the design of a code generator.

   b) Convert regular expression (a|b)*abb to NFA using Thompson's Construction and then perform NFA to DFA conversion. 7

6. a) What are the principal sources of optimization in compiler design? Explain with examples. 7

   b) Discuss Data-Flow Analysis in detail. How are Data-Flow Equations solved? 8

7. Write short notes on: **(Any two)** 2×5

   a) Mapping Values to Names

   b) Control Flow Graph

   c) Closure properties of REs

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POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2025
Programme: BE | Full Marks : 100
Course: Simulation and Modeling (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is simulation? Explain the steps of simulation study with flowchart. 8
   b) Why monte carlo method is best for computing static model? Use it to solve $\int_0^1 \frac{4}{1+x^2} dx$ using fifteen samples. 7
2. a) Explain the different types of time advancement mechanism. 7
   b) Explain human liver model with the help of necessary equations and draw analog computer. 8
3. a) What are different statements used in CSMP III? Explain interactive system. 8
   b) Explain Delayed call with the help of telephone Simulation. 7

OR

How utilization and occupancy are measured in discrete system simulation? Explain.

4. a) What are the steps involved in gathering statistics? Explain about Recording distribution and transit time. 8
   b) Define random numbers. What are the techniques to generate random numbers? 7

OR

Write down the steps to perform independent test using run up and run down test with necessary formulas. Why do we perform these tests?

5. a) Define pseudo random numbers. The following numbers have been generated 0.44, 0.19, 0.88, 0.27, 0.55, 0.13, 0.63, 0.74, 0.11 and 0.33. Use the Kolmogorov-Smirnov Test with a = 0.05 to determine, if the hypothesis is that the numbers are uniformly distributed on interval [0,1] can be rejected. [Use the critical value of D for a=0.05 and N=10 is 0.410.] 8
   b) Explain the organization of SIMSCRIPT. Write SIMSCRIPT program for bank queuing system. 7
6. a) Write GPSS code for manufacturing shop model when only one inspection is possible at a time with necessary assumption. 8
   b) Why is analysis of simulation output necessary? How is it done in replication of run? 7
7. Write short notes on: **(Any two)** 2×5
   a) Elimination of Initial bias
   b) Verification and Validation
   c) Cobweb Model

CamScanner

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2025
Programme: BE | Full Marks : 100
Course: Simulation and Modeling (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Define simulation and modeling. Explain the different component of the system. 7

   b) Explain how we can compute the given integral using Monte Carlo method. Use it to find the value of π. 8

2. a) Draw a Cobweb model to investigate a market in which the supply and demand functions are same. 7

   $D=12.4-1.2P$

   $S=1.0+0.9P_{-1}$

   $P_o=1.0$

   b) Consider a sample of 16 data points 0.15, 0.32, 0.48, 0.74, 0.22, 0.91, 0.63, 0.08, 0.55, 0.39, 0.27, 0.83, 0.17, 0.70, 0.44, 0.98. Perform a Kolmogorov-Smirnov (KS) test to determine if the sample follows a uniform distribution. Use a significance level of $\alpha=0.05$. 8

OR

What are the properties of random numbers? Generate seven 4-digit random numbers using the linear congruential generation method.

3. a) Explain Linear and nonlinear differential equation. Write CSMP III code for automobile suspension system. 7

   b) Write differential equation of automobile suspension wheel system and draw analog computer for it. 8

4. a) What are the various types of calls? Simulate the telephone system for delayed call system. 7

   b) What is discrete system simulation? How are utilization and occupancy measured in discrete system simulation? Explain with a suitable example. 8

OR

Differentiate Next event oriented and Fixed increment time advance mechanism.

5. a) Explain gap test and poker test to check the independence of random numbers. 7

   b) Explain the organization of SIMSCRIPT language with suitable example. 8

6. a) Write GPSS code for manufacturing shop model with necessary assumptions also draw the block diagram. 7

   b) Why is elimination of initial bias important? What are various nature of problems in simulation. 8

7. Write short notes on: **(Any two)** 2×5

   a) Steps of Simulation

   b) Real time Simulation

   c) Elimination of Initial Bias

CamScanner

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Simulation and Modeling (New)

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Define the term Simulation and modeling. What are the various principles of modeling? 7

   b) Draw the cobweb model in graph for: 8

   i. Fluctuation of market price.

   ii. Cobweb model for market economy graph from given data
   $D=12.4-1.2P$, $S=7.0-0.6$, P_{-1}, $D=S$, $P_0=1.0$

OR

What is a model? How system can be represented in various model? Explain with example.

2. a) Describe the Monte Carlo method. Integrate $\int_2^5 x^2\,dx$ using Monte Carlo simulation method using 10 samples. 8

   b) Differentiate between continuous and discrete system: Design an analog computer for automobile suspension wheel. 7

3. a) What is CSSL and why were they developed? Explain the structure of CSMP III. 8

   b) What are the various types of calls? Simulate telephone system for lost call system. 7

OR

What are the steps of simulation programming task? How can we simulate bank queue system? Explain.

4. a) What are the various components of a discrete system? Explain in detail. 7

   b) Explain the process of generation of Random number using LCG. Mention the properties of random numbers. 8

5. a) Using chi-square test with $\alpha = 0.05$, Check for the uniformity of the given random no. 8

0.87	0.69	0.12	0.83	0.60	0.95	0.01
0.93	0.27	0.58	0.23	0.99	0.25	0.19
0.28	0.19	0.88	0.36	0.89	0.33	0.68
0.31	0.35	0.49	0.64	0.91	0.05	0.23
0.41	0.43					

   b) Write GPSS code for manufacturing shop model when there are single inspection possible at a time. 7

6. a) Which section is used to declare the variables, entities and routines in SIMSCRIPT language? Declare variables entities and routines of telephone call simulation in the section. 8

   b) How does initial bias make an adverse effect? Discuss about the process of eliminating initial bias effect. 7

7. Write short notes on: **(Any two)** 2×5

   a) Next event time advance mechanism

   b) Utilization and occupancy

   c) Replication of run

CamScanner

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Networks (New)

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Define Computer Network. Explain the role of Internet Service Providers (ISPs) and Network Service Providers (NSPs). List the merits of client-server model over peer to peer model. 7
   b) Draw the TCP/IP model and explain one of the protocols of each layer. 8
2. a) Explain circuit switching mechanism. Explain the terms throughput, latency, bandwidth and jitter. 8
   b) What is flow control? Describe Stop-and-Wait, Go-Back-N, and Selective Repeat protocols with suitable diagrams. 7
3. a) Illustrate cyclic redundancy check method for error detection with an example. 8

   OR

   Explain the algorithm of CSMA/CD and draw the flowchart.

   b) An organization is planning to design 6 different branches for Marketing, IT, AI, Management, Accounting and Software. Each department consists of 28 computers with IP address192.108.254.0. Explain the process of allocating the IP address and subnet for each of the departments using above IP. 7
4. a) What is classful and classless address? Differentiate between link state and distance vector routing protocol. 7

   OR

   Explain IPv4 header format. Differentiate between IPv4 and IPv6.

   b) What are the function of transport layer. Explain the concept of socket programming in the transport layer protocol. 8
5. a) Explain the process of token based leaky bucket algorithm to address congestion during packet propagation. 5

   b) What are the roles of Domain Name Systems? Explain the working principles of Dynamic Host Configuration Protocol (DHCP). 5
   c) Explain the importance of proxy server and firewall in local area network. 5
6. a) What is data encryption? Explain the role of symmetric and asymmetric key cryptography with a suitable example. 8
   b) What do you mean by network Management? Explain SNMP protocol. 7
7. Write short notes on: **(Any two)** 2×5
   a) TCP congestion control
   b) NAT
   c) Switch versus router

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POKHARA UNIVERSITY

Level: Bachelor
Programme: BE.
Course: Computer Networks (New)

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Computer Network. Explain about Network Architecture. Discuss the criteria necessary for an effective and efficient network? 8
   b) Define protocol, stack and interface? Explain about protocol hierarchies in detail? 7
2. a) What is Transmission media. How ISDN interfaces and channel works? 7
   b) What is packet switching? How is this different from other switching techniques? 8
3. a) What is flow control? Explain different types of flow control mechanism. 7
   b) Illustrate checksum method for error control with an example. 8

OR

How does collision avoided during channel access? Mention the algorithm of CSMA/CD in detail.

4. a) Imagine yourself as a skillful computer network administrator. Given an IP address of 192.168.1. 0 / 25, your responsibility is to create four subnetworks for four departments namely A, B, C, D that require 99, 55, 29 and 13 hosts respectively. Find out the subnetwork address, broadcast address, and ranges of host addresses, subnetwork masks and wildcard masks for each department. 7
   b) Compare the header format of Internet Protocol version 4 (IPv4) with IPv6 and explain each fields in short. 8

OR

Write down the differences between adaptive and non-adaptive routing. Draw the scenario and explain Routing Information Protocol (RIP) with an example.

5. a) Explain the concept of socket programming in the transport layer protocol. 8
   b) What is congestion in a network? Compare between leaky bucket and token bucket algorithm with the operation how token bucket works. 7
6. a) What are DNS servers. Explain HTTP, FTP & proxy server. 8
   b) What do you mean by cryptography? Explain with algorithm Public vs Private Key Cryptography. 7
7. Write short notes on: **(Any two)** 2×5
   a) Firewall and digital certificate
   b) 4G Vs 5G
   c) Networking Hardware

CamScanner

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Computer Networks (New)

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) What is computer networking? Explain the peer-to-peer and client server architecture. 8
   b) What are protocols? Compare modern TCP/IP model with OSI reference model with neat diagram. 7
2. a) Explain Circuit Switching and Packet Switching with proper illustration. 8
   b) What are the functions of LLC and MAC sub layer? Explain framing in detail. 7

OR

Explain different flow control mechanism in Data link Layer in detail.

3. a) 7-bits even parity Hamming code is received as 1100010. Find the correct code that was transmitted. 8
   b) Explain IPV4 header format. Differentiate IPV6 with IPV4 in terms of features. 7

OR

Differentiate Distance Vector Routing and Link State Routing.
Explain OSPF Routing Protocol.

4. a) Design a network for Pokhara University having 4 departments having hosts 5, 16, 23, 27 such that IP address wastage in each department is minimum. Find out subnet mask, network address, broadcast address and wildcard mask and usable host range. The given IP is 10.10.10.0/24. 8
   b) Explain with an example the distance vector routing algorithm and compare it with link state routing. 7
5. a) What is congestion? How can congestion be controlled and managed? Explain with reference to leaky bucket algorithm. 8

   b) Explain the DHCP, FTP, HTTP and SMTP services of application layer. 7
6. a) What is data encryption and decryption? Explain the symmetric key cryptography for ensuring data security. 7
   b) "Networks must be protected from unauthorized access, attacks and breaches." Justify. 8
7. Write short notes on: **(Any two)** 2×5
   a) CSMA/CD
   b) Multiplexing and de-multiplexing technique in Transport Layer
   c) ISDN

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